

121. The total number of matrices

$$A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}, (x, y \in \mathbb{R}, x \neq y) \text{ for}$$

Which $A^T A = 3I_3$ is :

- (A) 6 (B) 3 (C) 4 (D) 2

$$\text{आव्यूहों } A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}, (x, y \in \mathbb{R}, x \neq y)$$

जिनके लिए $A^T A = 3I_3$ है, की कुल संख्या है :

- (A) 6 (B) 3 (C) 4 (D) 2

Ans. C

$$\text{Sol. } A^T A = \begin{bmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix} = 3I$$

$$= \begin{bmatrix} 8x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow 8x^2 = 3, 6y^2 = 3$$

$$x = \pm\sqrt{\frac{3}{8}}, y = \pm\sqrt{\frac{1}{2}}$$

Total combinations of $(x, y) = 2 \times 2 = 4$

122. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then determinant of A is:

- (A) 3 (B) 1/3 (C) 1/81 (D) 1/9

माना $A = [a_{ij}]$ तथा $B = [b_{ij}]$, 3×3 के दो वास्तविक आव्यूह इस प्रकार हैं कि $b_{ij} = (3)^{(i+j-2)}a_{ji}$, जहाँ $i, j = 1, 2, 3$. यदि B का सारणिक 81 है, तो A का सारणिक है :

- (A) 3 (B) 1/3 (C) 1/81 (D) 1/9

Ans. D

Sol. $B = \begin{bmatrix} a_{11} & 3a_{21} & 3^2 a_{31} \\ 3a_{12} & 3^2 a_{22} & 3^3 a_{32} \\ 3^2 a_{13} & 3^3 a_{23} & 3^4 a_{33} \end{bmatrix}$

$$|B| = 3^6 |A| = 81$$

$$|A| = \frac{1}{9}$$

123. Let $A = \{X = (x, y, z)^T : PX = 0 \text{ and } x^2 + y^2 + z^2 = 1\}$, where $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$, then the set A :

(A) contains more than two elements.

(B) contains exactly two elements.

(C) is a singleton.

(D) is an empty set.

यदि $A = \{X = (x, y, z)^T : PX = 0 \text{ तथा } x^2 + y^2 + z^2 = 1\}$ जबकि $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$ है, तो A :

(A) में दो से अधिक अवयव हैं।

(B) में मात्र दो अवयव हैं।

(C) एक ही अवयव वाला समुच्चय है।

(D) एक रिक्त समुच्चय है।

Ans. B

Sol. $|P| = (-3 + 36) - 2(2 + 4) + 1(-18 - 3)$
 $= 33 - 12 - 21 = 0$

$PX = 0$ has infinite solution

$$x + 2y + z = 0$$

$$-2x + 3y - 4z = 0$$

$$x + 9y - z = 0$$

put $z = \lambda$

$$x + 2y = -\lambda$$

$$\frac{x + 9y = \lambda}{y = 2\lambda / 7}$$

$$x = \frac{-11\lambda}{7}$$

$$(x, y, z) = \left(\frac{-11\lambda}{7}, \frac{2\lambda}{7}, \lambda \right)$$

$$x^2 + y^2 + z^2 = 1$$

$$\lambda^2 \left(\frac{121}{49} + \frac{4}{49} + 1 \right) = 1$$

$$\lambda = \pm \frac{7}{\sqrt{174}}$$

Two set of values

124. Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in \mathbf{R}$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _____

माना $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in \mathbf{R}$ तथा $A^4 = [a_{ij}]$ यदि $a_{11} = 109$ है, तो a_{22} बराबर है _____ ।

Ans. 10

Sol. $A^2 = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} x^2 + 1 & x \\ x & 1 \end{bmatrix}$

$$A^4 = \begin{bmatrix} x^2 + 1 & x \\ x & 1 \end{bmatrix} \begin{bmatrix} x^2 + 1 & x \\ x & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} (x^2 + 1)^2 + x^2 & \text{---} \\ \text{---} & x^2 + 1 \end{bmatrix} = \begin{bmatrix} 109 & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$(x^2 + 1)^2 + x^2 = 109 \Rightarrow x^2 = 9$$

$$a_{22} = x^2 + 1 = 10$$

125. Let A be a 3×3 matrix such that $\text{adj } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$.

If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair, $(|\lambda|, \mu)$ is equal to :

माना A एक 3×3 आव्यूह है, जिसके लिए $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ तथा $B = \text{adj}(\text{adj } A)$ हैं। यदि $|A| = \lambda$ तथा

$|(B^{-1})^T| = \mu$ हैं, तो क्रमित युग्म $(|\lambda|, \mu)$ बराबर है :

- (A) $(3, 81)$ (B) $\left(9, \frac{1}{9}\right)$ (C) $\left(3, \frac{1}{81}\right)$ (D) $\left(9, \frac{1}{81}\right)$

Ans. C

Sol. $|\text{adj } A| = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{vmatrix} = |A|^{n-1}$

$$\therefore |A|^2 = 9$$

$$|A| = \pm 3$$

$$|A| = \lambda = \pm 3$$

$$|\lambda| = 3$$

$$B = \text{adj}(\text{adj } A)$$

$$|B| = |\text{adj}(\text{adj } A)|$$

$$|B| = |A|^{(n-1)^2}$$

$$|B| = |A|^4$$

$$|B| = 81$$

$$\left| (B^{-1})^T \right| = \mu$$

$$|B^{-1}| = \mu$$

$$\frac{1}{|B|} = \mu$$

$$\mu = \frac{1}{81}$$

$$= \left(3, \frac{1}{81} \right)$$

126. If $A = \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$, $\left(\theta = \frac{\pi}{24} \right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $i = \sqrt{-1}$, then which one of the following is not true?

यदि $A = \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$, $\left(\theta = \frac{\pi}{24} \right)$ तथा $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, जबकि $i = \sqrt{-1}$ है, तो निम्न में से कौन-सा सत्य नहीं

है ?

(A) $a^2 - d^2 = 0$ (B) $a^2 - b^2 = \frac{1}{2}$ (C) $0 \leq a^2 + b^2 \leq 1$ (D) $a^2 - c^2 = 1$

Ans. B

Sol. $A^2 = \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$

$$= \begin{bmatrix} \cos^2 \theta + i^2 \sin^2 \theta & i(2 \cos \theta \sin \theta) \\ 2i \sin \theta \cos \theta & \cos^2 \theta + i^2 \sin^2 \theta \end{bmatrix} = \begin{bmatrix} \cos 2\theta & i \sin 2\theta \\ i \sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$A^3 = \begin{bmatrix} \cos 2\theta & i \sin 2\theta \\ i \sin 2\theta & \cos 2\theta \end{bmatrix} \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\theta & i \sin 3\theta \\ i \sin 3\theta & \cos 3\theta \end{bmatrix}$$

$$A^5 = \begin{bmatrix} \cos 5\theta & i \sin 5\theta \\ i \sin 5\theta & \cos 5\theta \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$a = \cos 5\theta \quad b = i \sin 5\theta$$

$$c = i \sin 5\theta \quad d = \cos 5\theta$$

$$a^2 - b^2 = \cos^2 5\theta - i^2 \sin^2 5\theta = \cos^2 5\theta + \sin^2 5\theta = 1$$

$$a^2 - d^2 = \cos^2 5\theta - \cos^2 5\theta = 0$$

$$a^2 + b^2 = \cos^2 5\theta - \sin^2 5\theta = \cos 10\theta = \cos \frac{5\pi}{12} \in (0, 1)$$

$$a^2 - c^2 = \cos^2 5\theta - i^2 \sin^2 5\theta = \cos^2 5\theta + \sin^2 5\theta = 1$$

127. Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, $Ax = b$ when the vector b on the right side is equal to b_1, b_2 and b_3 respectively. If

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \text{ and } b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}, \text{ then the determinant of } A \text{ is equal to :}$$

माना सदिश x_1, x_2 तथा x_3 , रैखिक समीकरण निकाय $Ax = b$ के हल हैं, जबकि दाईं ओर का सदिश b , क्रमशः b_1, b_2 तथा b_3 के बराबर है। यदि

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \text{ तथा } b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \text{ हैं, तो } A \text{ के सारणिक का मान है :}$$

(A) $\frac{3}{2}$

(B) 2

(C) 4

(D) $\frac{1}{2}$

Ans. B

Sol. Let $X = [x_1 \ x_2 \ x_3] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$$B = [b_1 \ b_2 \ b_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow |A| |X| = |B|$$

$$\Rightarrow |A| = \frac{|B|}{|X|} = \frac{4}{2} = 2$$

128. Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$. If $B = A + A^4$, then $\det(B)$:

माना $\theta = \frac{\pi}{5}$ तथा $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ है। यदि $B = A + A^4$, तो $\det(B)$:

(A) is one

(B) lies in (1, 2)

(C) lies in (2, 3)

(D) is zero

(A) is के बराबर

(B) अंतराल (1, 2) में है। (C) अंतराल (2, 3) में है। (D) zero के बराबर है।

Ans. B

Sol. $B = A(I + A^3)$

$$|B| = |A| |I + A^3|$$

$$|A| = 1$$

$$|B| = |I + A^3|$$

$$A^2 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$A^3 = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$$

$$A^3 + I = \begin{bmatrix} 1 + \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & 1 + \cos 3\theta \end{bmatrix}$$

$$|A^3 + I| = (1 + \cos 3\theta)^2 + \sin^2 3\theta$$

$$|B| = 1 + 2 \cos 3\theta + 1 = 2(1 + \cos 3\theta)$$

$$|B| = 2(1 + \cos 108^\circ) = 2(1 - \cos 72^\circ) = 2(1 - \sin 18^\circ)$$

$$= 2 \left(1 - \frac{\sqrt{5} - 1}{4} \right) = 2 \left(\frac{5 - \sqrt{5}}{4} \right) = \left(\frac{5 - \sqrt{5}}{2} \right)$$

- 129.** Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has :

- (A) no solution (B) exactly two solutions
(C) infinitely many solutions (D) a unique solution

माना A तथा B दो 3×3 वास्तविक आव्यूह हैं जबकि A सममित आव्यूह है तथा B विषम सममित आव्यूह है। तो रैखिक समीकरण निकाय, $(A^2B^2 - B^2A^2)X = O$, जबकि X एक 3×1 अज्ञात चरों का स्तम्भ आव्यूह है तथा O एक 3×1 शून्य आव्यूह है :

- (A) के ठीक दो हल हैं (B) का कोई भी हल नहीं है
(C) का केवल एक हल है (D) के अनन्त हल हैं

Ans. C

Sol. $(A^2B^2 - B^2A^2)^T = (A^2B^2)^T - (B^2A^2)^T$

$$= B^2A^2 - A^2B^2$$

$$\Rightarrow A^2B^2 - B^2A^2 \text{ is skew-symmetric}$$

$$\Rightarrow |A^2B^2 - B^2A^2| = 0$$

$$\Rightarrow \text{option (3)}$$

130. If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to _____.

यदि $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ तथा $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ हैं, तो $13(a^2 + b^2)$ बराबर है :

Ans. 13

Sol. $A = \begin{bmatrix} 0 & -\tan \theta \\ \tan \frac{\theta}{2} & 0 \end{bmatrix}$

$$(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$(I_2 + A) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} (I_2 - A)$$

$$\begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix} = \begin{bmatrix} a + b \tan \frac{\theta}{2} & a \tan \frac{\theta}{2} - b \\ b - a \tan \frac{\theta}{2} & b \tan \frac{\theta}{2} + a \end{bmatrix}$$

By comparison

$$a + b \tan \frac{\theta}{2} = 1 \quad \dots(1)$$

$$a \tan \frac{\theta}{2} - b = -\tan \frac{\theta}{2} \quad \dots(2)$$

$$b - a \tan \frac{\theta}{2} = \tan \frac{\theta}{2} \quad \dots(3)$$

$$b \tan \frac{\theta}{2} + a = 1 \quad \dots(4)$$

By, $(1)^2 + (2)^2$

$$a^2 + b^2 = 1$$

$$13(a^2 + b^2) = 13$$

- 131.** Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

माना A एक 2 कोटि का सममित आव्यूह है, जिसके अवयव पूर्णांक हैं। यदि A^2 के विकर्ण के अवयवों का योगफल 1 है, तो ऐसे आव्यूहों की संभावित संख्या है :

- (A) 1 (B) 12 (C) 6 (D) 4

Ans. D

Sol. Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$; $a, b, c, d \in \text{Integer}$.

$$A^2 = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ab + bc \\ ab + bc & b^2 + c^2 \end{bmatrix}$$

$$\text{Sum of diagonal entries} = a^2 + b^2 + b^2 + c^2 = 1$$

$$\Rightarrow a^2 + 2b^2 + c^2 = 1$$

$$\Rightarrow a = \pm 1, b = 0 = c$$

$$\text{OR } a = b = 0, c = \pm 1.$$

$$\Rightarrow 4 \text{ possible matrices}$$

- 132.** If $A = \begin{pmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{pmatrix}$ and $\det\left(A^2 - \frac{1}{2}I\right) = 0$, then a possible value of α is :

यदि $A = \begin{pmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{pmatrix}$ तथा $\det\left(A^2 - \frac{1}{2}I\right) = 0$ है, तो α का एक संभव मान है :

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. $A^2 = \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix} \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} \sin^2 \alpha & 0 \\ 0 & \sin^2 \alpha \end{bmatrix}$$

$$A^2 - \frac{I}{2} = \begin{bmatrix} \sin^2 \alpha - \frac{1}{2} & 0 \\ 0 & \sin^2 \alpha - \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\left(\sin^2 \alpha - \frac{1}{2} \right)^2 = 0$$

$$\sin \alpha = \frac{1}{\sqrt{2}}, \quad \sin \alpha = \frac{-1}{\sqrt{2}}$$

Rejected

$$\alpha = \frac{\pi}{4} \quad \alpha \in \left[0, \frac{\pi}{2} \right]$$

133. Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i = \sqrt{-1}$. Then the system of linear equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has:

(A) Infinitely many solutions

(B) Exactly two solutions

(C) A unique solution

(D) No solution

माना $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i = \sqrt{-1}$ हैं। तो रैखिक समीकरण निकाय $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$:

(A) का कोई नहीं है

(B) के अनंत हल हैं

(C) के मात्र दो हल हैं

(D) का अद्वितीय हल हैं

Ans. D

Sol. $A^2 = AA = \begin{bmatrix} i & i \\ -i & i \end{bmatrix} \begin{bmatrix} i & -i \\ -i & i \end{bmatrix} = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$$A^4 = A^2 A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$$

$$A^8 = A^4 A^4 = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} = \begin{bmatrix} 128 & -128 \\ -128 & 128 \end{bmatrix}$$

$$128x - 128y = 8$$

$$-128x + 128y = 64$$

system will have no solution

134. If A and B are two non-zero $n \times n$ matrices such that $A^2 + B = A^2 B$, then :

यदि A तथा B, $n \times n$ के दो शून्येत्तर आव्यूह इस प्रकार हैं कि $A^2 + B = A^2 B$ है, तो :

- (A) $A^2B = BA^2$ (B) $A^2 = I$ or $B = I$ (C) $AB = I$ (D) $A^2B = I$

Ans. A

Sol. $A^2 + B = A^2 B$

$$(A^2 - I)(B - I) \dots(1)$$

$$A^2 + B = A^2 B$$

$$A^2(B - I) = B$$

$$A^2 B (B - I)^{-1}$$

$$A^2 = B (A^2 - I)$$

$$A^2 = BA^2 - B$$

$$A^2 + B = BA^2$$

$$A^2B = BA^2$$

- 135.** Let α be a root of the equation $(a - c)x^2 + (b - a)x + (c - b) = 0$ where a, b, c are distinct real numbers such that

the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then, the value of $\frac{(a - c)^2}{(b - a)(c - b)} + \frac{(b - a)^2}{(a - c)(c - b)} + \frac{(c - b)^2}{(a - c)(b - a)}$ is :

माना समीकरण $(a - c)x^2 + (b - a)x + (c - b) = 0$, का मूल α है तथा आव्यूह जहाँ a, b, c भिन्न वास्तविक संख्याएँ हैं जबकि

आव्यूह $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ अव्युत्क्रमणीय है। तो $\frac{(a - c)^2}{(b - a)(c - b)} + \frac{(b - a)^2}{(a - c)(c - b)} + \frac{(c - b)^2}{(a - c)(b - a)}$ का मान है :

- (A) 3 (B) 12 (C) 6 (D) 9

Ans. A

Sol. $\Delta = 0 = \begin{vmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$

$$\Rightarrow \alpha^2(c - b) - \alpha(c - a) + (b - a) = 0$$

It is singular when $\alpha = 1$

$$\frac{(a - c)^2}{(b - a)(c - b)} + \frac{(b - a)^2}{(a - c)(c - b)} + \frac{(c - b)^2}{(a - c)(b - a)}$$

$$\frac{(a - b)^3 + (b - c)^3 + (c - a)^3}{(a - b)(b - c)(c - a)}$$

$$= 3 \frac{(a - b)(b - c)(c - a)}{(a - b)(b - c)(c - a)} = 3$$

136. The number of square matrices of order 5 with entries from the set $\{0, 1\}$, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is :

कोटि 5 के वर्ग आव्यूहों, जिनके अवयव समुच्चय $\{0, 1\}$ से हैं, प्रत्येक पंक्ति के सभी अवयवों का योग 1 है तथा प्रत्येक स्तम्भ के सभी अवयवों का योग भी 1 है, की संख्या है :

- (A) 225 (B) 125 (C) 150 (D) 120

Ans. D

Sol.

$$\begin{bmatrix} \text{-----} \\ \text{-----} \\ \text{-----} \\ \text{-----} \\ \text{-----} \end{bmatrix}$$

In each row and each column exactly one is to be placed :

$$\therefore \text{No. of such matrices} = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Alternate :

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} 5 \text{ways} \\ 4 \text{ways} \\ 3 \text{ways} \\ 4 \text{ways} \\ 1 \text{ways} \end{matrix}$$

Step-1 : Select any 1 place for 1's in row 1. Automatically some column will get filled with 0's.

Step-2 : From next now select 1 place for 1's. Automatically some column will get filled with 0's.

$\Rightarrow$ Each time one less place will be available for putting 1's.

Repeat step-2 till last row.

$$\text{Req. ways} = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

137. Let $x, y, z > 1$ and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$. Then $|\text{adj}(\text{adj} A^2)|$ is equal to :

माना $x, y, z > 1$ है तथा $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$ है। तो $|\text{adj}(\text{adj} A^2)|$ बराबर है :

- (A) 4^8 (B) 2^8 (C) 6^4 (D) 2^4

Ans. B

Sol. $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$

$$|A| = \frac{1}{\log x \log y \log z} \begin{vmatrix} \log x & \log y & \log z \\ \log x & 2\log y & \log z \\ \log x & \log y & 3\log z \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{vmatrix}$$

$$|A| = 2$$

$$\left| \text{adj}(\text{adj} A^2) \right| = |A|^8 = 2^8$$

138. Let $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where $i = \sqrt{-1}$

If $M = A^T B A$, then the inverse of the matrix $AM^{2023}A^T$ is

माना $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ -3 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$ है, जहाँ $i = \sqrt{-1}$ है।

यदि $M = A^T B A$ है, तो आव्यूह $AM^{2023}A^T$ का व्युत्क्रम है

(A) $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

Ans. C

Sol. $AA^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ -3 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{-3}{\sqrt{10}} \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$B^2 = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2i \\ 0 & 1 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 1 & -3i \\ 0 & 1 \end{bmatrix}$$

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$$B^{2023} = \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$$

$$M = A^T B A$$

$$M^2 = M \cdot M = A^T B A A^T B A = A^T B^2 A$$

$$M^3 = M^2 \cdot M = A^T B^2 A A^T B A = A^T B^3 A$$

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$$M^{2023} = \dots\dots\dots A^T B^{2023} A$$

$$A M^{2023} A^T = A A^T B^{2023} A A^T = B^{2023}$$

$$= \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$$

$$\text{Inverse of } (A M^{2023} A^T) \text{ is } \begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$$

139. Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13} B^{26} - B^{26} A^{13}$ is symmetric

(S2) $A^{26} C^{13} - C^{13} A^{26}$ is symmetric

Then

(A) Both S1 and S2 are true

(B) Only S1 is true

(C) Both S1 and S2 are false

(D) Only S2 is true

माना 3×3 के आव्यूहों A, B, C में A सममित है तथा B और C विषम सममित है। कथनों

(S1) $A^{13}B^{26} - B^{26}A^{13}$ सममित है

(S2) $A^{26}C^{13} - C^{13}A^{26}$ सममित है

का विचार कीजिए। तो

(A) S1 तथा S2 दोनों सत्य है

(B) केवल S1 सत्य है

(C) S1 तथा S2 दोनों असत्य है

(D) केवल S2 सत्य है

Ans. D

Sol. $S_1 : (A^{13}B^{26} - B^{26}A^{13})^T = (A^{13}B^{26})^T - (B^{26}A^{13})^T$

$$\Rightarrow (B^{26})^T (A^{13})^T - (A^{13})^T (B^{26})^T$$

$$\Rightarrow (BT)^{26} (A^T)^{13} - (A^T)^{13} (B^T)^{26}$$

$$\Rightarrow (-B)^{26} A^{13} - A^{13} (-B)^{26}$$

$$\Rightarrow -(A^{13}B^{26} - B^{26}A^{13}) \text{ so skew symmetric}$$

$$S_2 : (A^{26}C^{13} - C^{13}A^{26})^T$$

$$= (C^T)^{13} (A^T)^{26} - (A^T)^{26} (C^T)^{13}$$

$$= -C^{13}A^{26} + A^{26}C^{13}$$

So, symmetric matrix

Hence S_2 is true and S_1 is false

140. Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then

माना α व β वास्तविक संख्याएं हैं। एक 3×3 आव्यूह A है लिए $A^2 = 3A + \alpha I$ है। यदि $A^4 = 21A + \beta I$ है, तब

(A) $\beta = 8$

(B) $\beta = -8$

(C) $\alpha = 4$

(D) $\alpha = 1$

Ans. B

Sol. $A^2 = 3A + \alpha I$

$$A^3 = 3A^2 + \alpha A$$

$$A^3 = 3(3A + \alpha I) + \alpha A$$

$$A^3 = 9A + \alpha A + 3\alpha I$$

$$A^4 = (9 + \alpha)A^2 + 3\alpha A$$

$$= (9 + \alpha)(3A + \alpha I) + 3\alpha A$$

$$= A(27 + 6\alpha) + \alpha(9 + \alpha)$$

$$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$$

$$\Rightarrow \beta = \alpha(9 + \alpha) = -8$$